

LESSON 315 (1.5 AATD)

Lesson Objective

During this lesson, VOR, and localizer holding patterns will be introduced.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

Lesson Content (Simulator)

Consider starting with little or zero wind to allow the student to focus on holding procedures.

System Power Up

- Student Tells, Student Does.
 - Have the student start the simulator using the checklist and assist.
 - Ensure that the student checks GPS database currency.
 - Half-left, half-up, flag out of view.
 - Ensure that there are no red 'x's.
 - Have the student start the engine using the checklist.

Instrument Cockpit Check

- Student Tells, Student Does.
 - Have the student read each item from the checklist out loud.
 - Demonstrate to the student what needs to be done for each item.
 - Discipline the student to set bugs, bearing pointers, and any auxiliary features.

Before Taxi Checks

- Student Tells, Student Does.
 - Airspeed indicator (PFD and standby): zero.
 - Attitude indicator (PFD): pitch/bank deviations +/- 5 degrees.
 - Standby attitude indicator: discuss standby battery check and adjust the miniature aircraft (cannot be performed in the ATD).
 - Altimeter (PFD and standby): +/- 75 feet of station elevation.
 - Vertical speed indicator (PFD): reads 0 or use current indication.
 - Trim coordinator: wings level or no vector when still; inclinometer centered when still; skids when turning.
 - HSI: matches compass and turns appropriately.
 - Magnetic compass: indicates heading properly, no bubbles, card present, and swings freely.

Give the student a departure via radar vectors. The instructor simulates ATC. Have the student switch to the appropriate frequencies throughout the simulator lesson.

VOR Holding Patterns

- Standard hold:
- Instructor Tells, Student Does.
 - If necessary, pause the simulator while giving holding instructions.
 - Start with a direct entry.
 - Have the student track/intercept radials or reposition them to be south of the VOR.
 - Give the student instructions to fly directly to the MLB VOR.
 - Show the student two different methods to fly to the VOR:
 - Pressing OBS and flying the radial to the VOR. Ensure the ground track is on the radial.
 - Using the bearing pointer to fly to the VOR. This is the preferred method, which allows the CDI to be twisted to the inbound leg for the hold.
 - Give the student a standard holding instruction (omit saying right turns).
 - *Hold south of the MLB VOR on the 180 radial, EFC (add 15 min to clock on MFD).*
 - From the current position (with the sim paused), ask the student if they can determine the entry.
 - Once entry is understood, proceed with the 6 Ts:
 - Turn - upon crossing fix, turn to the outbound heading.
 - Talk - advise ATC established in the hold.
 - Time - start the timer when the fix is crossed or wings level, whichever happens last.
 - Twist - twist to the inbound course.
 - Throttle - adjust throttle for 90 kts.
 - Track - after one minute or the holding distance, turn inbound and track the inbound course.
 - Ensure the student is making proper wind and time corrections.
 - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
 - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
 - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
 - When holding, ensure the student is continuing with the radial scan.
 - Remind the student about the cone of confusion within 1 mile of the VOR and to not make big corrections.
 - Repeat procedures with different instructions until the student is proficient.

- Non-standard hold:
- Instructor Tells, Student Does.
 - In the simulator, drag the aircraft south of the TRV (Treasure) VOR.
 - Give the student instructions to fly directly to the TRV VOR.
 - Tune and identify the station. Verify that the HSI/CDI is set up to the proper radio (VLOC1/VLOC2).
 - If necessary, pause the simulator while giving holding instructions. Give either a teardrop or parallel instruction.
 - *Hold northeast of the TRV VOR on the 020 radial, **left turns**, EFC (add 15 min to clock on MFD).*
 - From the current position (with the sim paused), ask the student if they can determine the entry.
 - Once entry is understood, unpause the sim and proceed with the 6T's:
 - Parallel:
 - Turn - upon crossing fix, turn to the outbound heading.
 - Time - start a timer for one minute.
 - Twist - twist to the inbound course.
 - Track - after 1 minute, make a 225° turn to track the inbound course.
 - As a non-standard hold, it will be a right 225° turn.
 - Throttle - adjust throttle for 90 kts.
 - Talk - advise ATC established in the hold.
 - Teardrop:
 - Turn - After crossing the fix, offset the outbound heading by 30 degrees.
 - Non-Standard holds: subtract 30° from outbound heading.
 - Remember: **LARS** (Left add, Right subtract).
 - Time - start a timer for 1 minute.
 - Twist - twist to the inbound course.
 - Track - after 1 minute, make a standard rate turn to join the inbound leg.
 - As a non-standard hold, it will be a left turn.
 - Throttle - adjust throttle for 90 kts.
 - Talk - advise ATC established in the hold.
 - Ensure the student is making proper wind and time corrections.
 - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
 - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
 - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.

- When holding, ensure the student is still continuing with the radial scan.
- Repeat procedures with different instructions until the student is proficient.

Localizer Holding Patterns:

- Standard hold:
 - If necessary, pause the simulator while giving holding instructions.
 - Give the vectors to intercept the ILS 9R or Loc BC 27L at KMLB.
 - Ensure the student tune & ID's the localizer frequency.
 - On the ILS 9R, give the student instructions to hold at FIBOP or on a DME fix from the MLB VOR due to weather delay/traffic congestion at KMLB.
 - On the ILS 9R: *TECH FLIGHT 636, Hold west of FIBOP on the MLB localizer, EFC (add 15 min to clock on MFD).*
 - On the LOC BC 27L: *TECH FLIGHT 636, Hold east of CAPEN on the MLB localizer, EFC (add 15 min to clock on MFD).*
 - Once entry is understood, unpause the sim and proceed with the 6 Ts:
 - Turn: to the outbound heading upon crossing the fix.
 - Time: Start the timer when wings level or when crossing the fix, whichever happens **last**.
 - Twist: Advise the student that for a localizer hold you **do not "twist"** the inbound leg. for eg. The MLB localizer stays at a 094 course.
 - Track.
 - Throttle: set.
 - Talk (Advise ATC when crossing the fix when established on the inbound leg).
 - Ensure the student is making proper wind and time corrections.
 - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
 - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
 - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
 - When holding, ensure the student is still continuing with the radial scan.
 - Repeat procedures with different instructions until the student is proficient.
- Non-standard hold:
 - If necessary, pause the simulator while giving holding instructions.
 - Repeat steps mentioned above, drag the airplane on the simulator to bring the student back on the localizer.
 - On the localizer BC 27L: *"TECH FLIGHT 636, Hold at CAPEN as published, EFC (add 15 min to clock on MFD).*

- On the ILS 9R: *TECH FLIGHT 636, Hold west of FIBOP on the MLB localizer, left turns, EFC (add 15 min to clock on MFD).*
- Once entry is understood, unpause the sim and proceed with the 6T's:
 - Turn: to the outbound heading upon crossing the fix.
 - Time: Start the timer when wings level or when crossing the fix, whichever happens **last**.
 - Twist: Advise the student that for a localizer hold you **do not "twist"** the inbound leg. for eg. The MLB localizer stays at a 094 course.
 - Track.
 - Throttle: set.
 - Talk (Advise ATC when crossing the fix when established on the inbound leg).
- Ensure the student is making proper wind and time corrections.
 - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
 - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
 - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
- When holding, ensure the student is still continuing with the radial scan.
- Repeat procedures with different instructions until the student is proficient.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 316.
- Assign Homework:
 - Assign holding practice assignment.
 - Study holding entries and procedures.

Completion Standards

The student should demonstrate the correct procedures in order to execute VOR and localizer holding patterns.